

Enhanced Clinical Image Captioning via Vision-Language Model Augmentation

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Abstract

This project aims to improve automatic report generation in the X-ray domain using an integrated model that combines computer vision and natural language algorithms. The proposed model is designed to interpret X-ray images and generate descriptive reports, mimicking a specialist's interpretation.

Multiple models are explored, starting from a simple model and progressively moving to more complex ones. The baseline model extracts handcrafted features from images and uses them to generate reports using GPT-2. Following that, deep learning methods are employed to extract image features. These models include DenseNet121, Vision Transformers (ViT) and ResNet-50. A hybrid model is then developed to fuse deep and handcrafted features. In addition to these proposed models, multiple fine-tuning strategies are explored for the ViT + GPT-2 architecture, including frozen, full fine-tuning, and LoRA approaches.

The quality of the generated reports is evaluated at the word, phrase, and semantic levels using BLEU, ROUGE-L, and BERTScore. The best-performing model (LoRA End-to-End (ViT + GPT-2)) with a **BLEU-1 score of 0.3518, ROUGE-L of 0.2937, and BERTScore of 0.6364.**

1. Introduction

X-ray images play a vital role in detecting various diseases. However, interpreting and analysing these images manually by specialists can be complex, time-consuming and require high-level expertise. Moreover, the process is susceptible to subjectivity in interpretations among experts. To address these challenges, many research efforts have explored the possibility of automating this task using deep learning and artificial intelligence techniques and making the process more accurate, efficient, and consistent.

Numerous studies and algorithms have been developed in this field, beginning with classical methods that rely on the extraction of handcrafted features. This was later followed by deep learning approaches like Convolutional Neural Networks (CNNs), which extract local features hierarchically from images. More recently, Transformers have made a significant breakthrough in both natural language processing (NLP) — particularly in text generation — and in computer vision, where they are used for image feature extraction[1]. Unlike CNNs, where each part of the image is processed locally, Transformers use self-attention to model the relationships between all parts of the input, whether it be words in a text or regions in an image.

2. Literature review

The task of automatically generating reports to describe X-ray images in a way that closely resembles the work of specialists is considered challenging. It requires skills and expertise in several areas, including computer vision and natural language processing. There is Many previous studies and research efforts have addressed this topic and attempted to improve the quality of generated reports. Some relied on classical methods, others on deep learning approaches and the use of transformers—whether for extracting features from images or report generation—and some combined different methods to achieve the best possible results.

2.1 Handcrafted Features Extraction

Handcrafted features are extracted from images, capturing specific visual patterns such as edges, texture, brightness, sharpness, and statistical properties of pixel values. Examples include the Sobel filter detects sharp transitions such as bone-tissue boundaries [2], while histogram brightness helps identify lung density variation[3]. Moreover, texture features derived from the Grey Level Co-occurrence Matrix (GLCM) differentiate between healthy and diseased lungs[4], among many additional handcrafted features. These features are interpretable and understandable by humans, in contrast to the features learned by deep learning methods. They can also serve as complementary information to deep features, potentially improving performance.

2.2 CNNs and Vision Transformers (ViT) for Feature Extraction

ResNet-50 (ImageNet)

CNNs extract hierarchical features: low-level features like edges and textures, mid-level features like patterns and curves, and high-level features like abstract shapes and structures. They are characterised by their ability to extract spatial patterns that are generally difficult to obtain using handcrafted features[1].

One commonly used CNN is a pre-trained ResNet-50 (trained on ImageNet). It is typically used to extract features from images after removing the classification layer from the network. These features represent the global visual semantics of the X-ray. Although ResNet-50 is not trained specifically on X-ray images, it is effective because many of the low-level and mid-level features are general and shared across different types of images, such as edges and textures, in other

words, the information obtained from general images is adapted to be useful in the domain of X-rays [5]

DenseNet-121 (CheXpert)

The DenseNet-121 model is pre-trained on a large dataset of 200,000 chest X-ray images known as the CheXpert dataset. It is used to extract 1024 global features from each image [6] [7]. The main difference between CheXpert DenseNet-121 and ResNet-50 (described in the previous section) is the domain of training data. DenseNet-121 is specifically trained on chest X-rays, not on natural images (like dogs, cars, etc.). Therefore, it is better adapted to medical imaging tasks and is expected to perform better in this context.

Vision Transformers (ViT)

ViT encoders use self-attention mechanisms to learn long-range dependencies between different regions of the image, effectively capturing the global context in medical images such as chest X-rays. Pretrained models like ViT-base-patch16-224-in21k trained on ImageNet-21k have proven to be a promising approach in the field of computer vision and medical imaging[8] [1].

2.3 Decoder for report generation and Retrieval-Augmented Generation

Large language models for medical report generation have been explored in recent studies. One approach employs the GPT-2 language model with cross-attention to generate reports by integrating textual decoding with visual embeddings in a multimodal captioning framework [1]. To further improve the generated report, retrieval-based augmentation is used, where semantically relevant past reports are retrieved from the ChromaDB vector database. These retrieved findings are ranked by semantic similarity using SentenceTransformer, and the top matching report is concatenated with the generated output to produce more informed, contextually enriched reports. This approach helps add natural clinical phrasing, fills in gaps missed by the GPT-2 model, and achieves better domain alignment in generated text, much like how a doctor looks at similar past cases to write a better report [1].

2.4 Summary and Project Objectives

This project aims to further enhance report generation quality by exploring models that combine multiple approaches, including handcrafted features, DenseNet-121, and ViT architectures. Moreover, multiple fine-tuning strategies will be investigated, such as LoRA and full fine-tuning. The proposed method also includes hyperparameter tuning and data preprocessing techniques.

3. Methodology

The proposed methodology involves a sequence of steps. Starting with preprocessing the data, including both images and text. Then, features are extracted from the images using classical and deep learning methods. After that, multiple models are built and evaluated to generate medical reports by learning from the reference reports and the extracted image features. Finally, the generated reports for the test images are assessed using several evaluation methods, such as BLEU and BERTScore. The overview of the proposed methodology is shown in Figure 1.

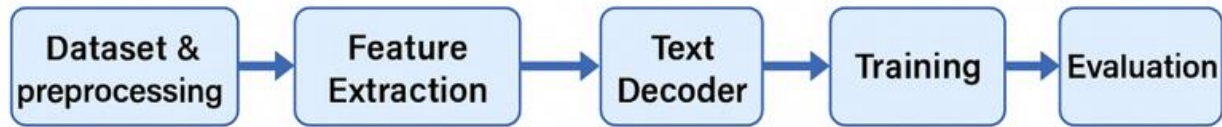


Figure 1: Overview of the Proposed Methodology

3.1 Dataset & Preprocessing

The dataset used in this study is the Open-I Chest X-ray collection from Indiana University. It comprises chest X-ray images along with corresponding patient reports. Each patient has from 1 to 5 X-ray images, and the reports are linked to the images through a unique ID.

The dataset was first explored using Exploratory Data Analysis (EDA) to understand its content. The data was then prepared through several steps:

- Firstly, missing values were filled in key columns such as Impression, Findings, Indication, and Comparison.
- Secondly, a cleaning function was defined to remove irrelevant patterns such as HTML, tags, bullet numbers, placeholders (e.g., XXXXX), URLs, and non-letter characters, as well as to expand English contractions (e.g., "won't" → "will not").
- Thirdly, the final report was constructed by merging the Impression, Findings, and Indication columns, with redundancy removed when present in the concatenated text.
- Finally, each patient's X-ray image was linked to its corresponding report using the unique UID field.

After preprocessing, A total of **3,851 image–report pairs**, samples were obtained for use in the project, with each sample consisting of a chest X-ray image and a corresponding patient report.

3.2 Image Feature Extraction and Preprocessing

This step involved extracting both handcrafted and deep learning-based features:

1. Handcrafted Features

A total of approximately 76 handcrafted features will be extracted for each image. After extraction, normalisation is applied to ensure they are compatible with the deep learning features since they will be combined later. Table 1 provides a summary of extracted handcrafted features used in Chest X-ray analysis.

Table 1: Summary of Handcrafted Visual Features Used in Chest X-ray Analysis

Feature Name	Description	Output Dim	Benefit	Ref
Sobel	Average gradient magnitude using Sobel filters (edge strength).	1	Detects sharp transitions such as bone-tissue boundaries.	[2]
Brightness	Mean of the grayscale histogram (brightness level).	1	Indicates density-related pathologies like pneumonia.	[3]
Gray-Level Co-occurrence Matrix(GLCM)	Texture props from GLCM (contrast, correlation, energy, homogeneity, dissimilarity).	5	• Differentiates between healthy and diseased lungs. • Detects structural lung issues.	[4]
Local Binary Pattern(LBP)	Histogram of Local Binary Pattern (captures texture).	59	• Powerful at detecting fine-grained patterns in lung tissue (sometimes hard to detect by eye). • Diagnosing complex radiographic cases	[9]
Histogram_stats	Statistical features: mean, std, skewness, kurtosis, entropy.	5	• Captures distributional properties across the whole image. • Separate localised vs. diffuse abnormalities.	[10]
Canny	Edge density using Canny edge detector (percentage of edge pixels).	1	Flag the presence of abnormal edges or foreign objects.	[11]
Laplacian	used to estimate image sharpness.	1	• Distinguishes blurry vs. sharp images • Useful for focus assessment • Ensuring pathology clarity.	[12]
Brightness range	Difference between max and min brightness (intensity range).	1	• Detects contrast variation, useful in identifying hazy pathology. • Exposure quality. • Visibility assessment	[3]

2. Deep Features

ResNet-50 (ImageNet)

As mentioned in the literature review, deep features are extracted using a pre-trained ResNet-50 model. It is used to extract features from images after removing the classification layer from the network. CNN extract hierarchical features progressing from low-level to high-level from X-ray images. In the early CNN layers, features like rib edges, spine lines, and lung boundaries are extracted; in the mid layers, features like lung fields, costophrenic angles, and opacities are extracted; and in the deep layers, features like lung shapes and heart contours are extracted. The final feature vector is of size 2048 per image.

DenseNet-121 (CheXpert)

As mentioned in the literature review, the DenseNet-121 model is pre-trained on a large dataset of 200,000 chest X-ray images known as the CheXpert dataset. The DenseNet-121 extracts 1024 global features from each image.

Vision Transformers (ViT)

As discussed in the literature review, ViT captures the global context in medical images. In this project, deep features were extracted using a pre-trained ViT-base-patch16-224-in21k model, which was trained on ImageNet-21k. The images are first resized to 224×224 pixels, then divided into 14×14 non-overlapping patches. Finally, a 768-dimensional [CLS] token embedding is extracted to summarise the image, reflecting its semantic content.

3.3 Text Decoder

There are several sizes of GPT-2 that differ in the number of parameters and, consequently, in performance. These include GPT-2 Small, Medium, Large, and XL. The appropriate decoder is typically chosen based on the task complexity and available computational resources. In this project, the small version of GPT-2 was used due to limited resources.

As discussed in the literature review, the GPT-2 language model with cross-attention is used to generate reports. It integrates textual decoding with visual embeddings in a multimodal captioning framework, allowing the model to attend to image features while generating the report. To align dimensions, a linear projection layer is used to map the fused image feature vector of any input size to 768 dimensions to match the GPT-2 hidden size.

3.4 Model Architecture

This section presents the different model configurations and fine-tuning strategies evaluated in the project.

3.4.1 Model Variants

Several models will be evaluated, starting with a simple model that uses handcrafted features. Then, multiple deep learning models will be tested, including **DenseNet-121** (pre-trained on chest X-rays), **Vision Transformer (ViT)**, and **ResNet-50** (both pre-trained on general image datasets). Finally, hybrid models will be explored by combining deep and classical methods in order to improve overall performance.



Figure 2: Model Variants Across Stages

After choosing the model that achieved the best performance, additional hyperparameter tuning was performed to further improve the results. For example, the number of epochs, learning rate, and batch size were adjusted to optimise training.

3.4.2 Fine-Tuning Strategies for ViT + GPT-2

To compare the Vision Transformer (ViT) model with existing work in [1], multiple fine-tuning strategies were explored.

- **Stage 1:** The ViT encoder was frozen, and the GPT-2 decoder was fully fine-tuned.
- **Stage 2:** The ViT encoder remained frozen, while the GPT-2 decoder was fine-tuned using the parameter-efficient LoRA method.
- **Stage 3:** Both the ViT encoder and GPT-2 decoder were jointly fine-tuned using LoRA, (end-to-end training)

The performance of all strategies was compared to decide which one performs better in terms of enhancing generalisation, reducing overfitting, and lowering computational cost, especially when working with a small dataset (see **Figure 3**).

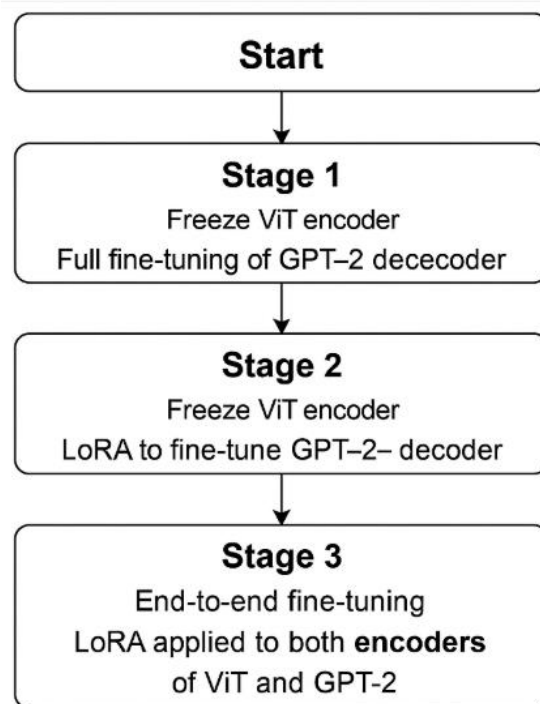


Figure 3: Overview of the Fine-Tuning Strategies Applied to the ViT-GPT-2 Architecture.

Note: Other visual encoders (e.g., DenseNet, ResNet), mentioned earlier, were used in **frozen mode only**, without any fine-tuning to evaluate their strength in extracting features without additional training. The **projection layer** that maps visual features to the GPT-2 hidden space was always **trained**.

3.5 Training and Inference

- The dataset is prepared such that each sample contains a tokenized caption and a projected image feature vector. The dataset is then split into 70% training, 15% validation, and 15% testing subsets.
- The model is trained to predict the next token based on previous tokens in an auto-regressive manner using cross-entropy loss and the image features via cross-attention.
- Training runs for up to 10 epochs (20 in some cases), with early stopping if validation loss does not improve for 4 consecutive epochs. The best-performing model is saved.
- Model's performance is monitored using both training and validation loss, as well as perplexity.
- During inference, the GPT-2 model is used to generate reports based only on the projected image features. As a clarification, the true caption is not available to the model during inference.
- The generated captions are decoded into readable text with a maximum sequence length of 70 tokens to ensure concise and relevant outputs. A beam search with 3 beams is used to improve report quality.

3.6 Evaluation

For evaluation, a comparison between the ground-truth captions and the model's predictions is conducted. The quality of the generated reports is evaluated using a combination of BLEU, ROUGE-L, and BERTScore metrics. This combination allows for multiple levels of evaluation.

- Word-level accuracy using BLEU,
- Phrase-level alignment using ROUGE-L, and
- Semantic-level accuracy using BERTScore.

4. Result and discussion:

This section shows the experimental findings and analyzes the performance of different model configurations and fine-tuning strategies evaluated in this project.

4.1 Results by Model Variants

As outlined in the methodology, a sequence of models was evaluated across four stages, ranging from a handcrafted baseline to advanced fusion approaches. In all models, the feature extraction components (e.g., DenseNet, ResNet, ViT) were used in **frozen mode without fine-tuning**, while **the projection layer and GPT-2 decoder were fine-tuned**. The figures below show the results.

- 4.1.1 Model 0: Handcrafted-only

As shown in Figure 4, the training loss and perplexity steadily decrease throughout all epochs. In contrast, the validation loss and perplexity show a sharp decline up to epoch 4, followed by a slower decrease until epoch 8. Beyond this point, both metrics begin to rise, indicating signs of overfitting.

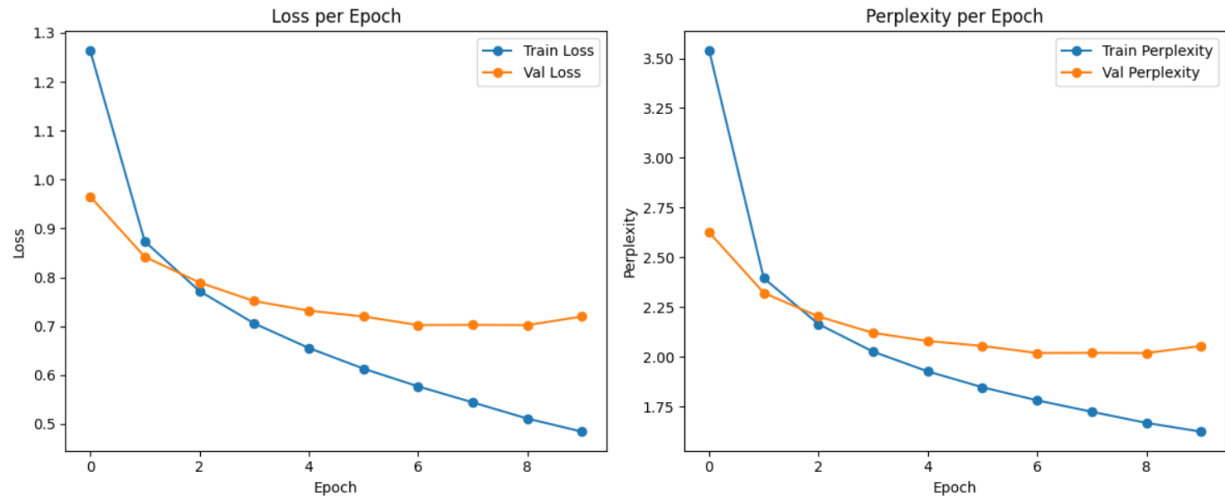


Figure 4: Training and validation loss (left) and perplexity (right) curves for the **Handcrafted-only model**.

- 4.1.2 Model 1a: DenseNet-121 (CheXpert)

As shown in Figure 5, the training loss and perplexity steadily decrease throughout all epochs. The validation loss and perplexity show a sharp decline up to epoch 4, followed by a slower but stable improvement. Unlike the Handcrafted-only model, there is a small increase in the validation metrics at the last epoch, suggesting a reduction in the overfitting.

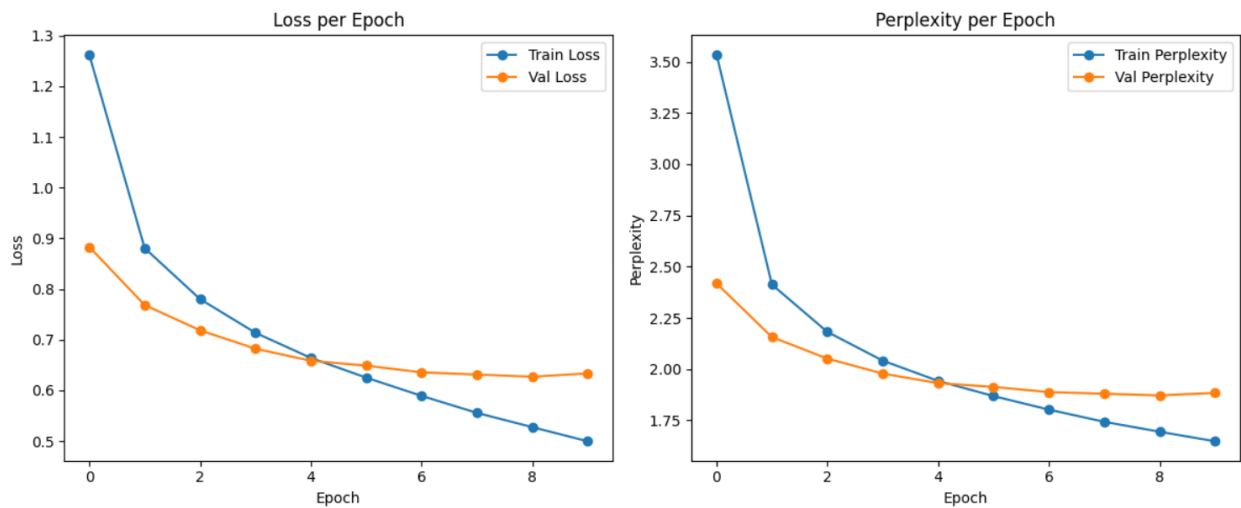


Figure 5: Training and validation loss (left) and perplexity (right) curves for the **DenseNet-121 (CheXpert) model**

- 4.1.3 Model 1b: ViT

As shown in Figure 6, the training loss and perplexity steadily decrease throughout all epochs. The validation loss and perplexity show a sharp decline up to epoch 3, followed by a slower decrease that plateaus after epoch 6. There is no noticeable increase in the validation metrics at later epochs, suggesting a reduction in the overfitting compared to the previous models. However, the ViT model shows a larger gap between training and validation loss/perplexity, indicating slower learning.

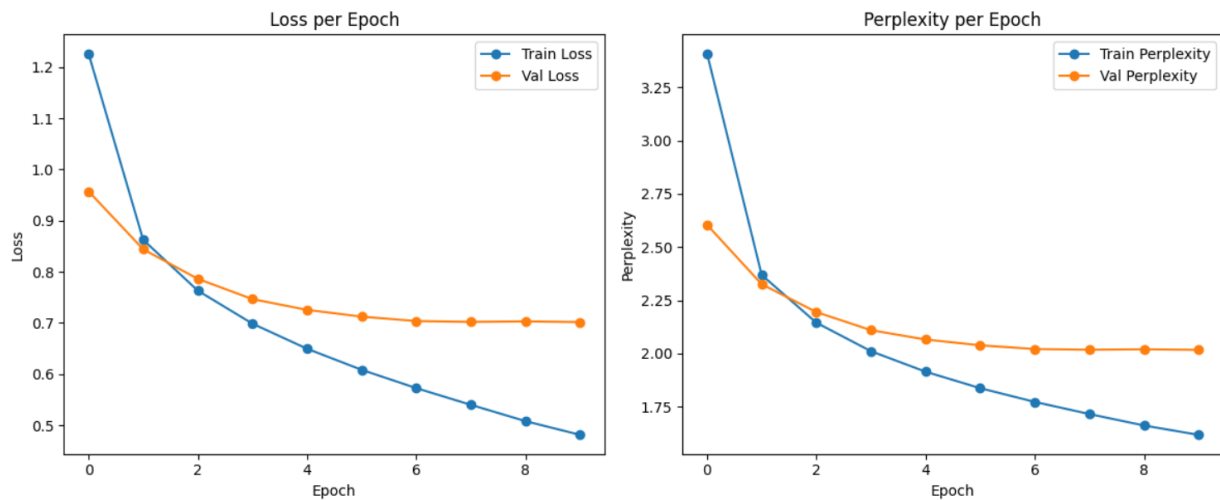


Figure 6: Training and validation loss (left) and perplexity (right) curves for the **ViT model**

- 4.1.4 Model 1c: ResNet-50

Figure 7 shows performance that is similar to and close to the previous models, with no additional points to discuss.

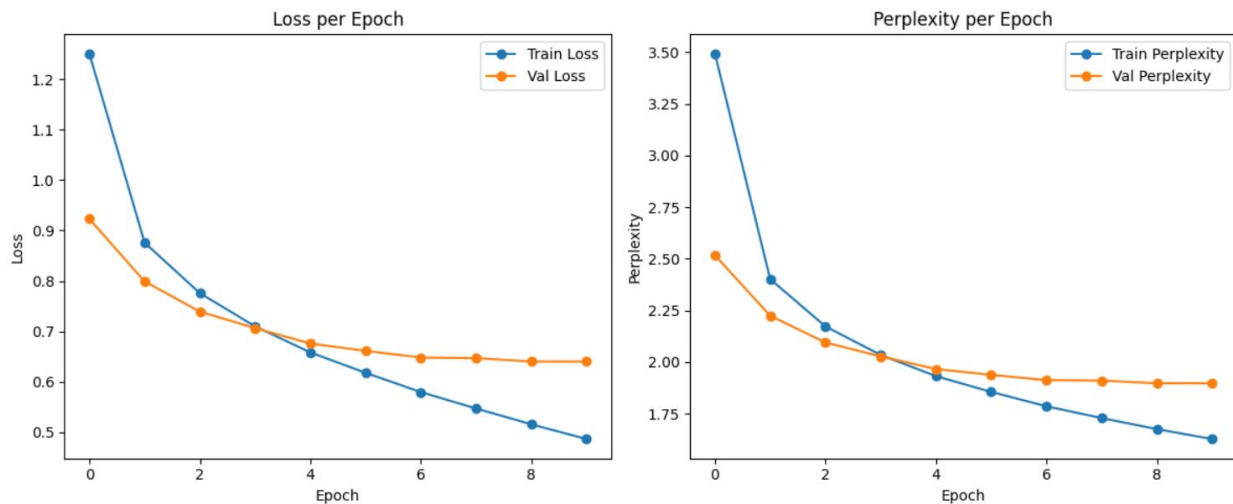


Figure 7: Training and validation loss (left) and perplexity (right) curves for the **ResNet-50 model**

- 4.1.5 Model 2a: DenseNet + Handcrafted

The combined model in Figure 8 shows slightly more stable validation metrics in later epochs compared to each model, suggesting improved generalisation.

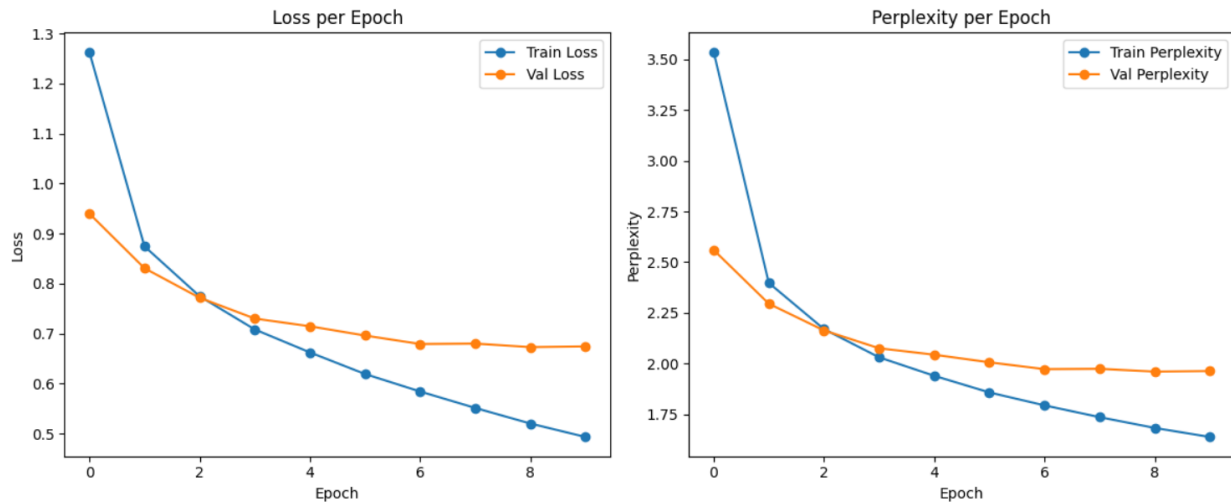


Figure 8: Training and validation loss (left) and perplexity (right) curves for the **DenseNet + Handcrafted model**

- 4.1.6 Model 2b: ViT + Handcrafted

Figure 9 suggests that adding handcrafted features does not noticeably enhance ViT's ability to generate better reports.

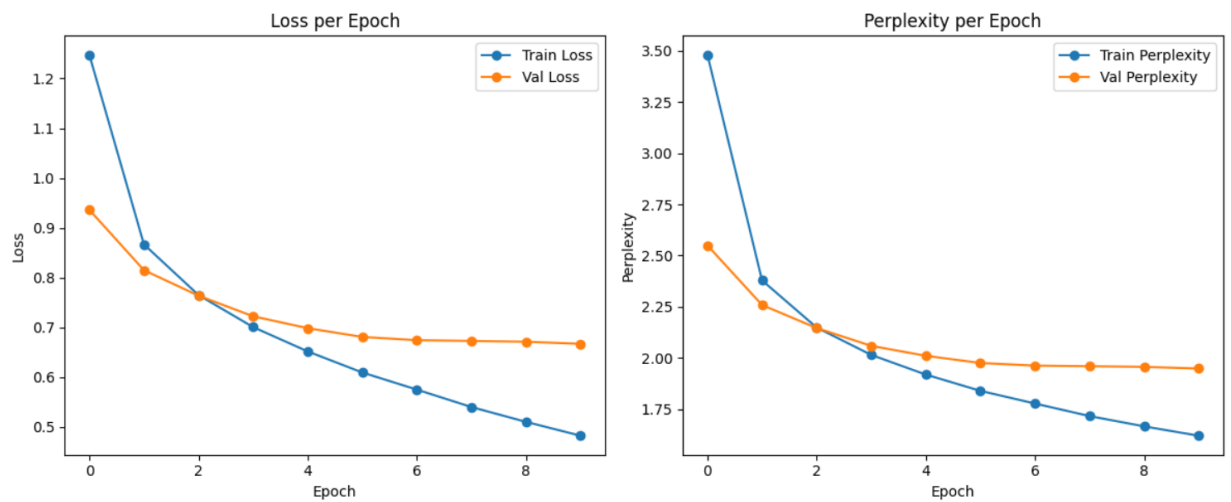


Figure 9: Training and validation loss (left) and perplexity (right) curves for the **ViT + Handcrafted model**

- 4.1.7 Model 3: Full Fusion (e.g., DenseNet + ViT + handcrafted)
Figure 10 shows no breakthrough in convergence or generalisation. However, the final validation loss is slightly lower than other models

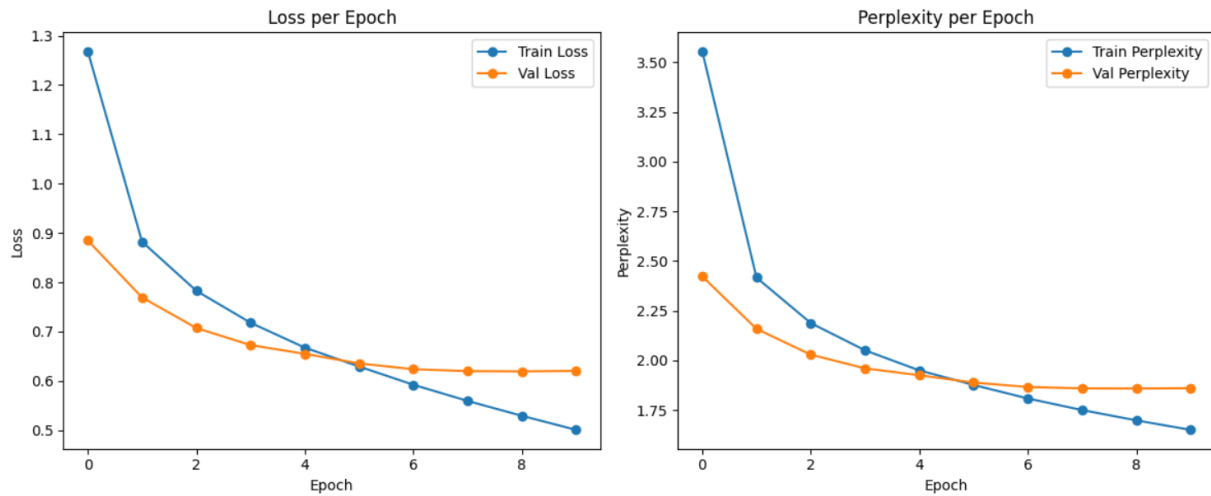


Figure 10: Training and validation loss (left) and perplexity (right) curves for **the DenseNet + ViT + handcrafted model**

4.1 Results by Fine-Tuning Strategy for ViT + GPT-2

This stage explored different training settings and parameter-efficient methods (e.g., LoRA vs. full fine-tuning) to further enhance performance and generalization.

- 4.2.1 Stage 1: Full GPT-2 tuning (ViT frozen)

As shown in Figure 11, the training loss and perplexity steadily decrease across all epochs. In contrast, the validation loss and perplexity drop initially until around epoch 6, after which they gradually rise, indicating the onset of overfitting due to limited data and a frozen ViT encoder.

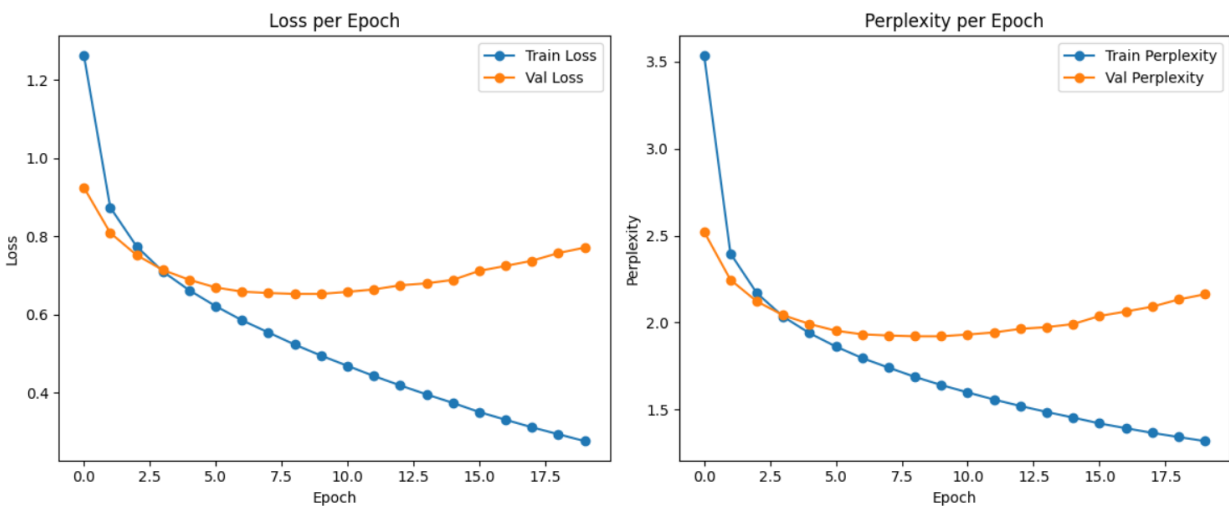


Figure 11: Training and validation loss (left) and perplexity (right) curves for the **full GPT-2 fine-tuning with the ViT encoder frozen**.

- 4.2.2 Stage 2: LoRA on the decoder

As shown in Figure 12, both training and validation loss drop sharply in the first few epochs and stabilize quickly. Similarly, perplexity rapidly decreases and then flattens out, indicating that applying LoRA on the decoder helps the model converge efficiently without clear signs of overfitting.

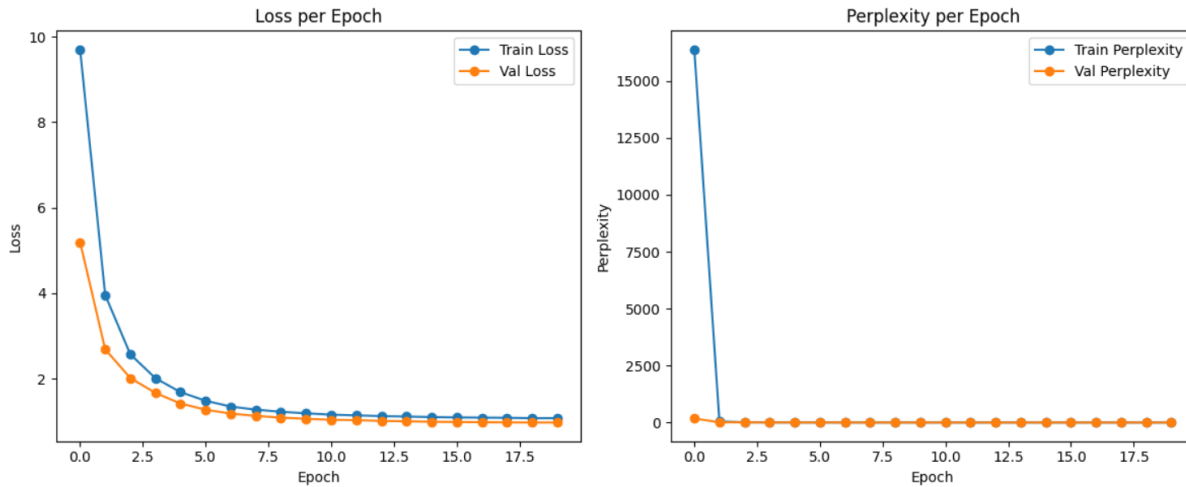


Figure 12: Training and validation loss (left) and perplexity (right) curves for the **LoRA GPT-2 fine-tuning with the ViT encoder frozen**.

- 4.2.3 Stage 3: LoRA on both (end-to-end)

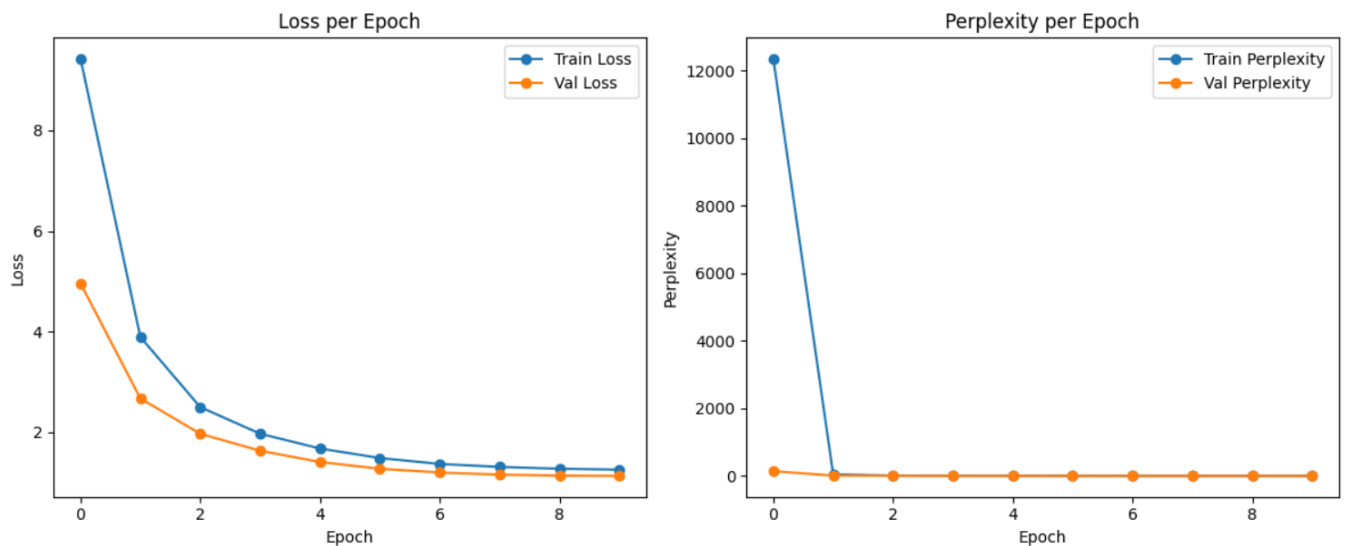


Figure 13: Training and validation loss (left) and perplexity (right) curves for the LoRA on both (end-to-end)

Compare validation loss, perplexity, BLEU/ROUGE/BERTScore across proposed models and with Prior Work

Table 2: Evaluation results of all models on the test set

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L	BERT-Score (F1)
Handcrafted-only + Full Fine-Tuning GPT-2	0.2939	0.1790	0.1224	0.0823	0.2648	0.5594
Frozen DenseNet-121+ Full Fine-Tuning GPT-2	0.2611	0.1746	0.1218	0.0789	0.2915	0.5856
Frozen ViT+ Full Fine-Tuning GPT-2	0.2960	0.1837	0.1253	0.0814	0.2730	0.5682
Frozen ResNet-50+ Full Fine-Tuning GPT-2	0.2401	0.1382	0.0904	0.0594	0.2460	0.5583
Frozen DenseNet + Handcrafted + Full Fine-Tuning GPT-2	0.2937	0.1801	0.1242	0.0854	0.2753	0.5717
Frozen ViT + Handcrafted + Full Fine-Tuning GPT-2	0.2892	0.1779	0.1207	0.0772	0.2761	0.5683
Frozen (DenseNet + ViT) + handcrafted++ Full Fine-Tuning GPT-2	0.3007	0.1848	0.1277	0.0878	0.2644	0.5560
ViGPT2 (Paper Baseline)	0.571	0.385	0.291	0.226	0.433	-
LoRA Decoder Only (ViT + GPT-2)	0.3288	0.2058	0.1369	0.0888	0.2839	0.5752
Full Fine-Tuning — Decoder Only (GPT-2)	0.2951	0.1781	0.1205	0.0802	0.2583	0.5532
LoRA End-to-End (ViT + GPT-2)	0.3518	0.2265	0.1544	0.1029	0.2937	0.6364

Performance Trends and Discussion

- LoRA End-to-End (ViT + GPT-2) achieves the highest scores.
- Model 1a (DenseNet121 pretrained on CheXpert) achieves the best BERTScore, reflecting its strong semantic understanding.
- Model 2a (DenseNet121 + Handcrafted features) outperforms both its components (Model 0 and Model 1a) individually across most metrics. This demonstrates that handcrafted features complement the deep model.
- In the report generation field in the medical field, BERTScore is a more reliable evaluation method than others due to its ability to assess semantic similarity.
- The batch size for training data was tuned as well. Generally, larger batch sizes (e.g., 16) produced better generalization than Smaller batches (e.g., 2 or 4)

5. Conclusion & Future Work

In this project, a multi-stage chest X-ray report generation framework was evaluated. Started with a baseline handcrafted model in Stage 0, followed by many deep learning models, including DenseNet121, ViT, and ResNet-50 in Stage 1, then fusion models with handcrafted features in Stage 2, and finally a full fusion model with augmentation in Stage 3. In addition to the architectural stages, multiple fine-tuning strategies were explored for ViT + GPT-2: frozen, full fine-tuning, and LoRA.

Several ideas can be investigated and explored to enhance model performance and generalization. For example, using GPT-2 Medium or even GPT-3 if computational resources allow. Additionally, applying sampling methods during inference (top-k or nucleus) for better variation. Adding medical-style prompt tuning may produce more structured outputs. Finally, using image data augmentation like flipping or adding noise to improve the model's generalization.

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